

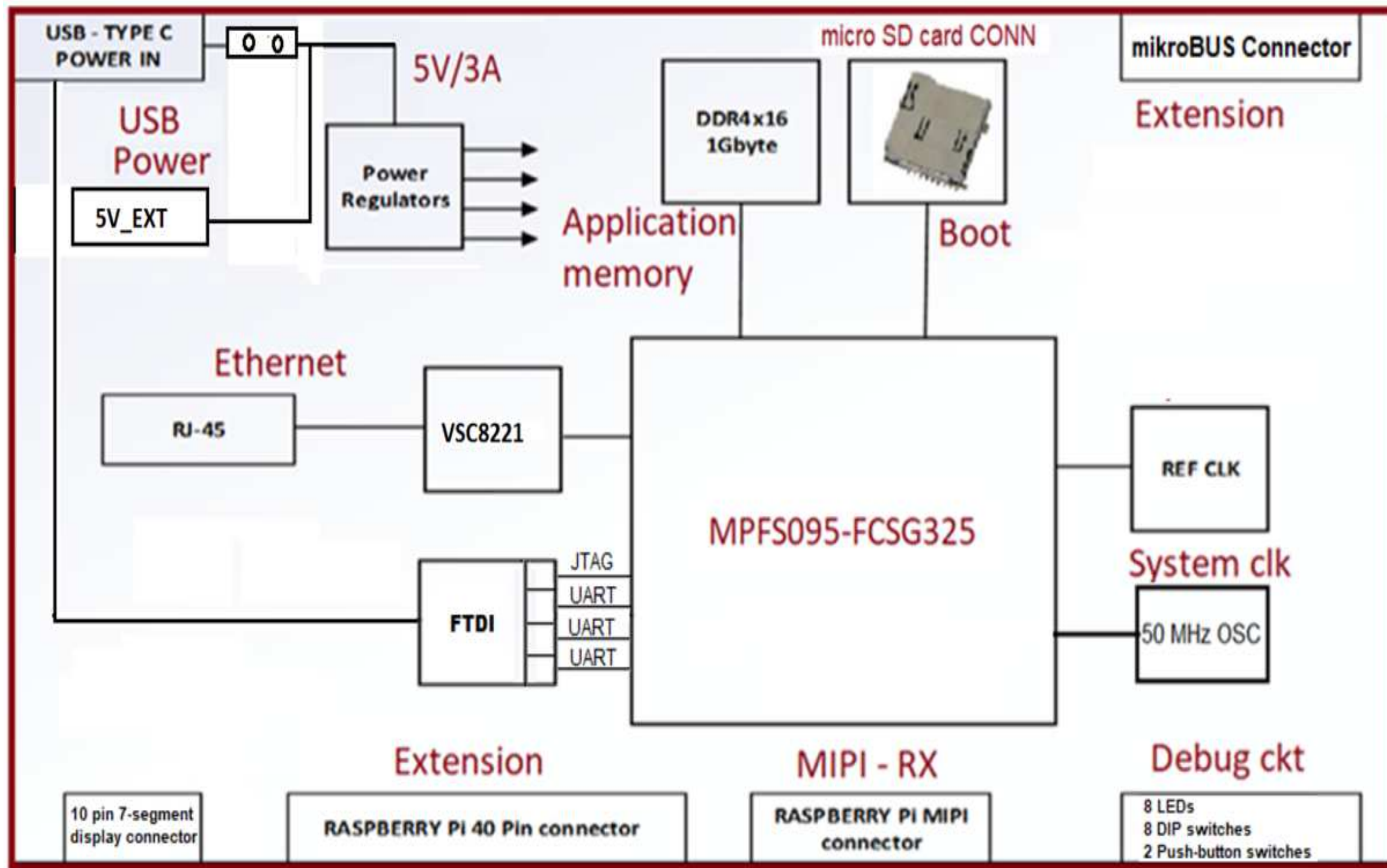
PolarFire SoC Discovery Kit

Part Number: MPFS-DISCO-KIT

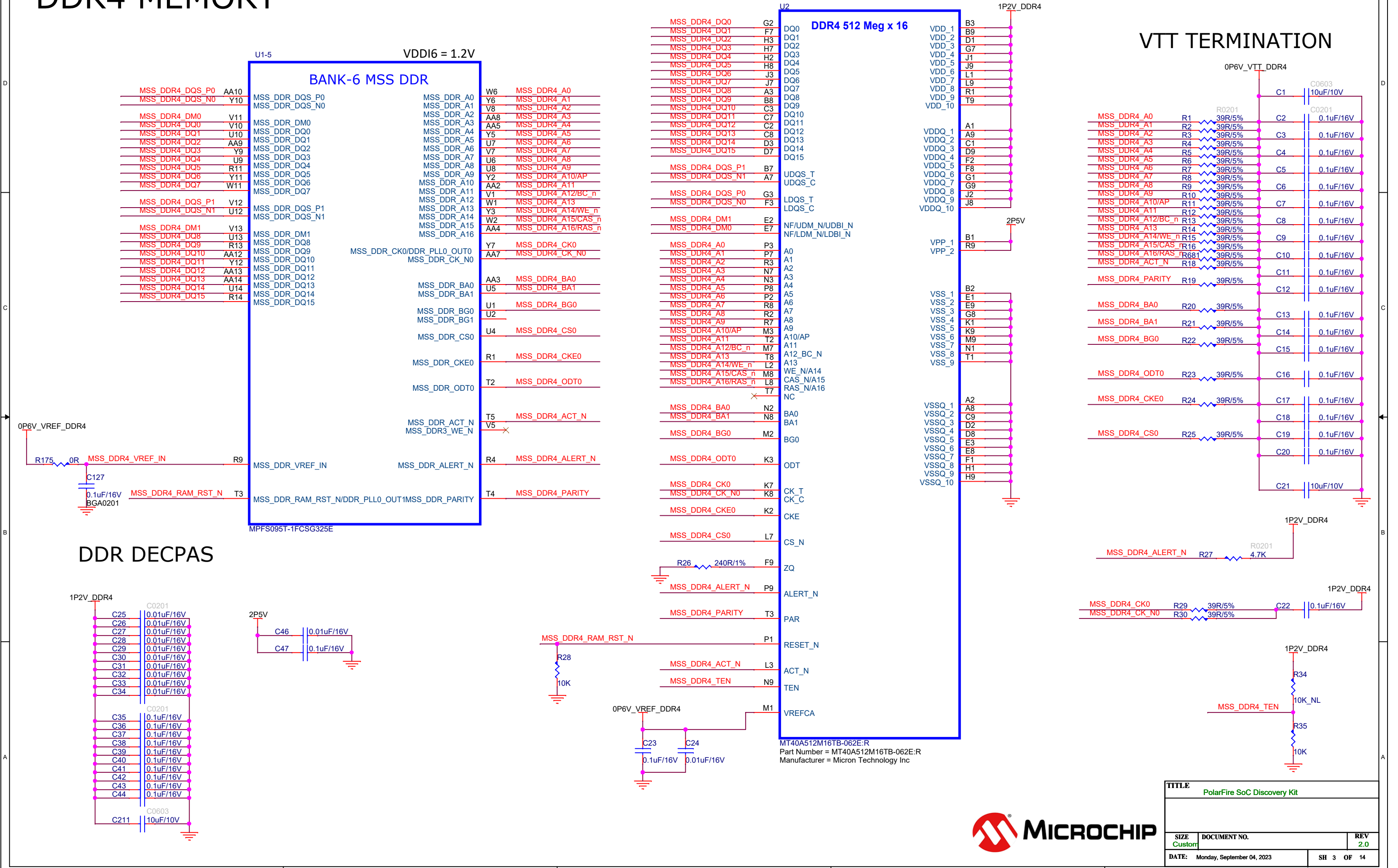
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Project Name:			PolarFire SoC Discovery Kit		
Rev	Part Number:	DVP-100-000578-002		PACTRON:	
2.0					
Originator:					
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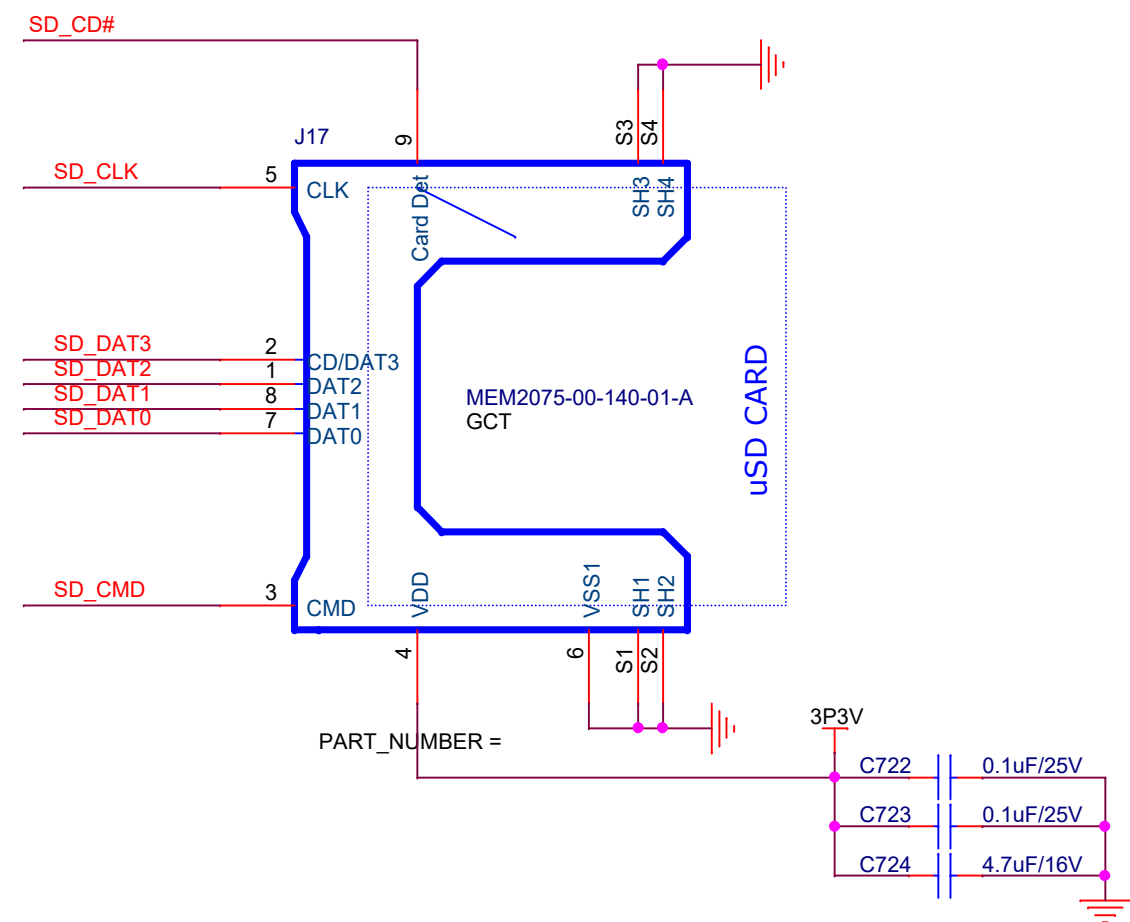
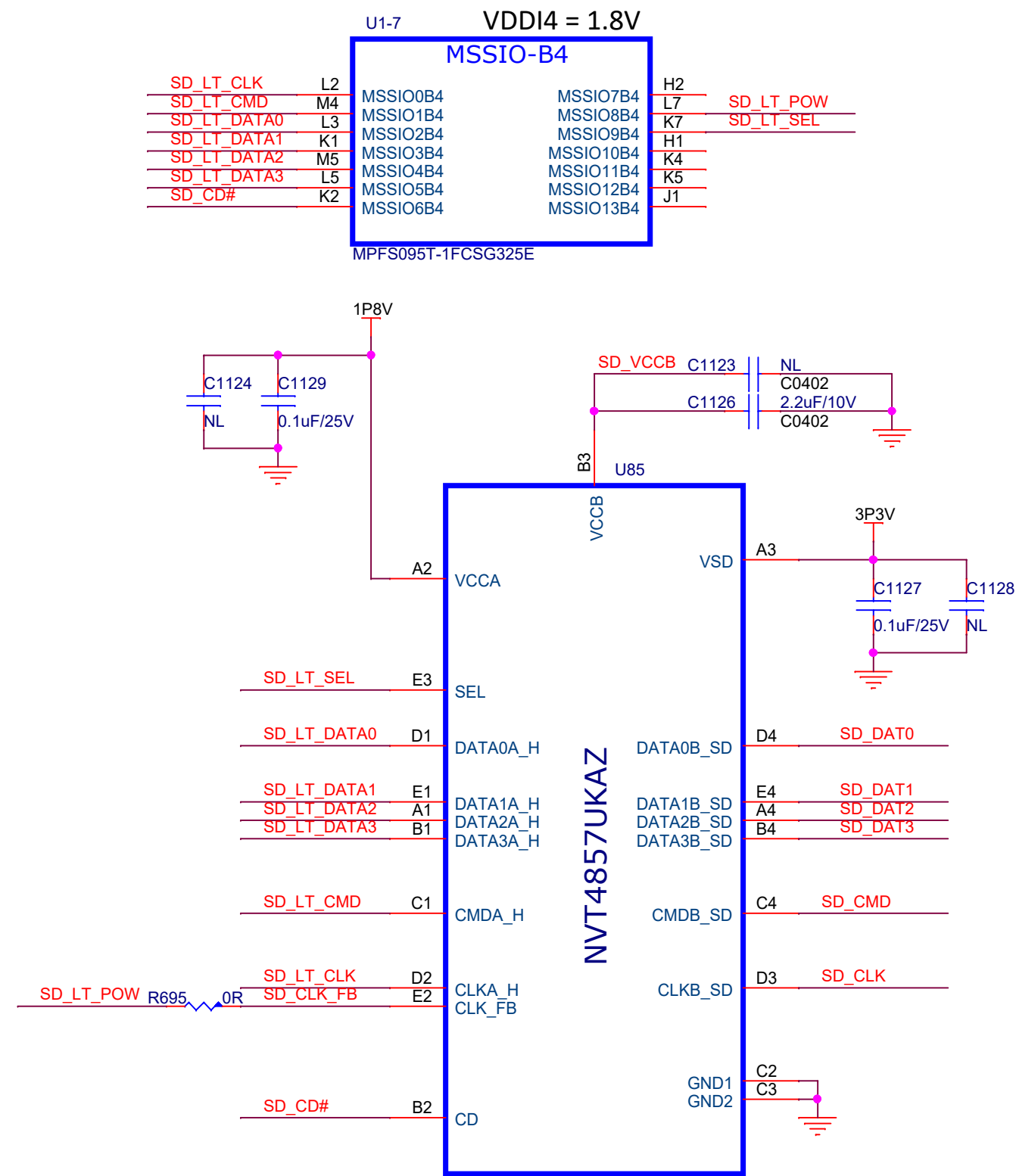


DDR4 MEMORY



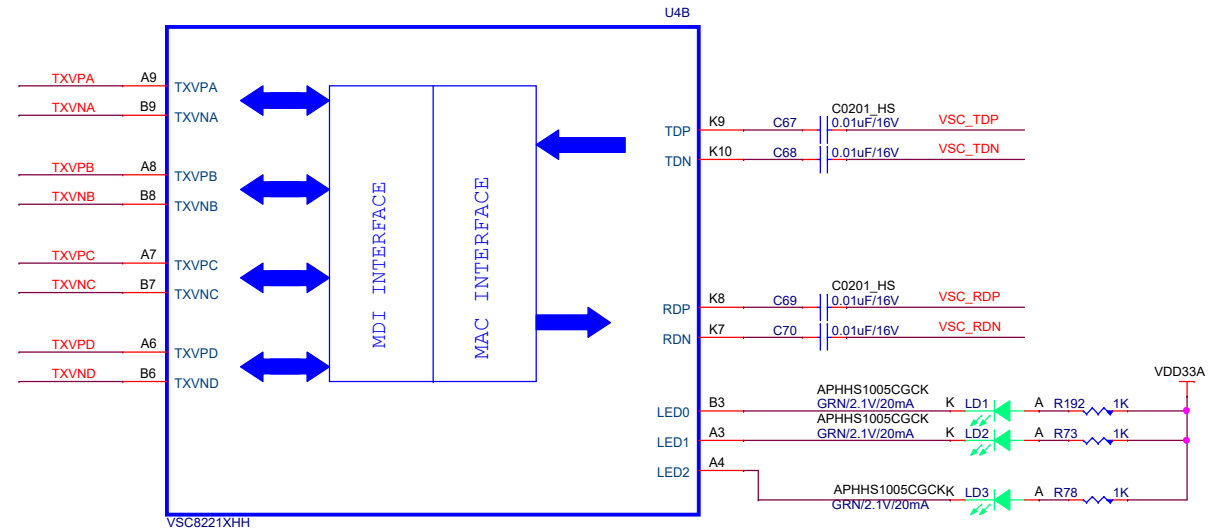
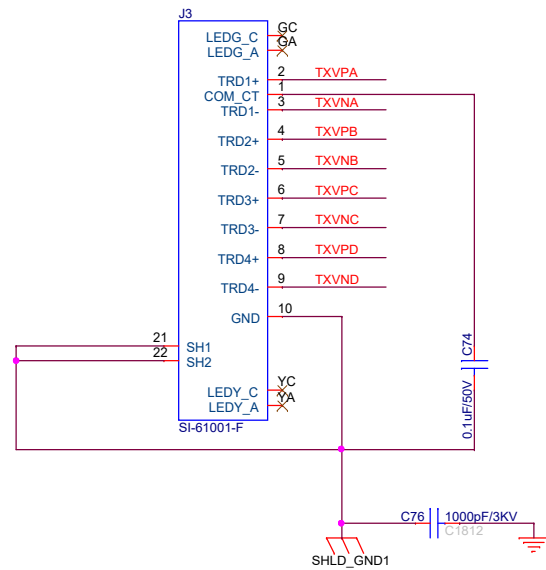
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BANK4- uSD Conneciton

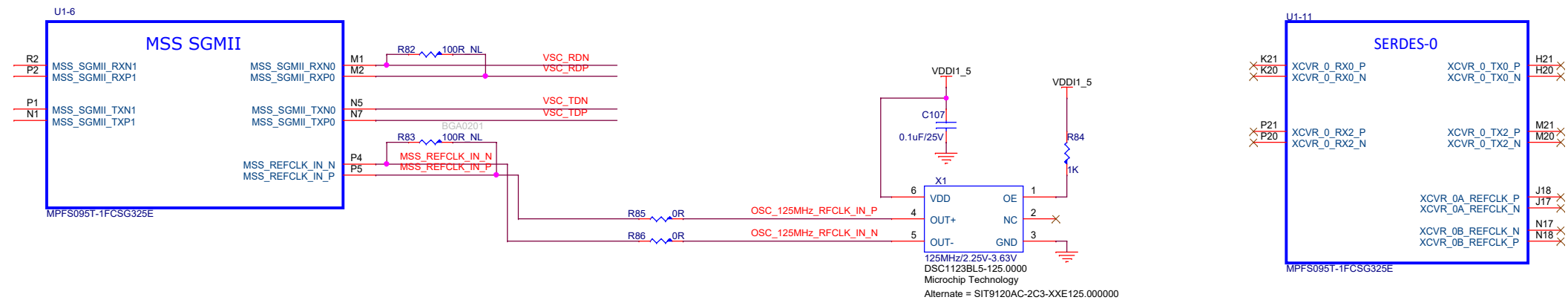
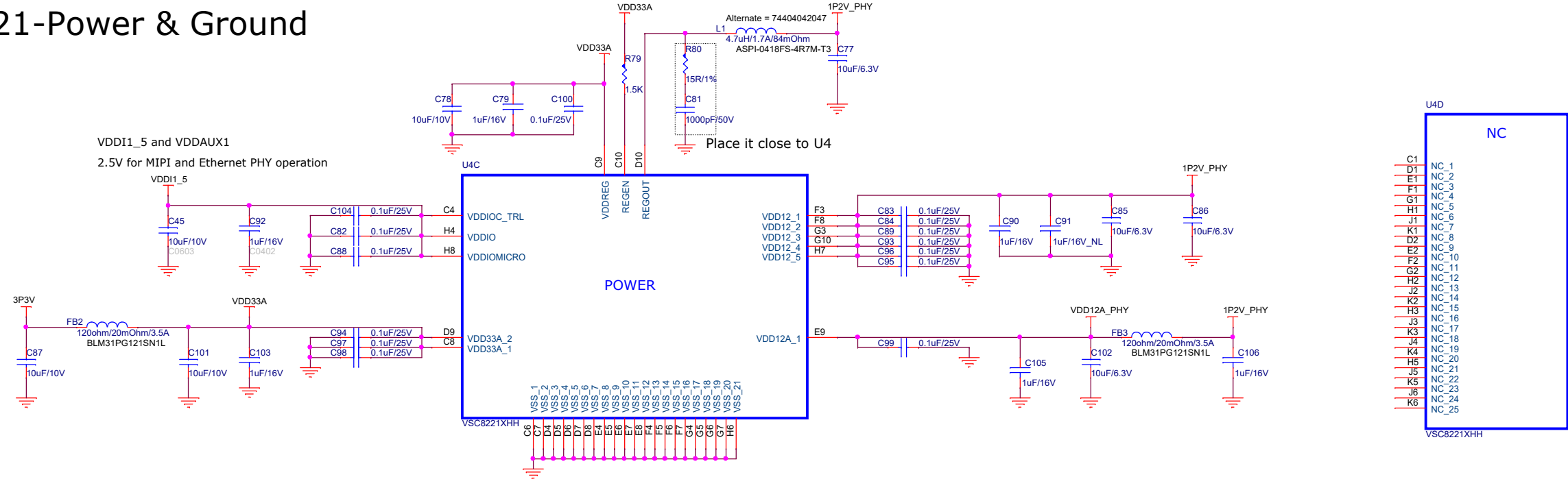


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VSC8221 - SGMII Interface

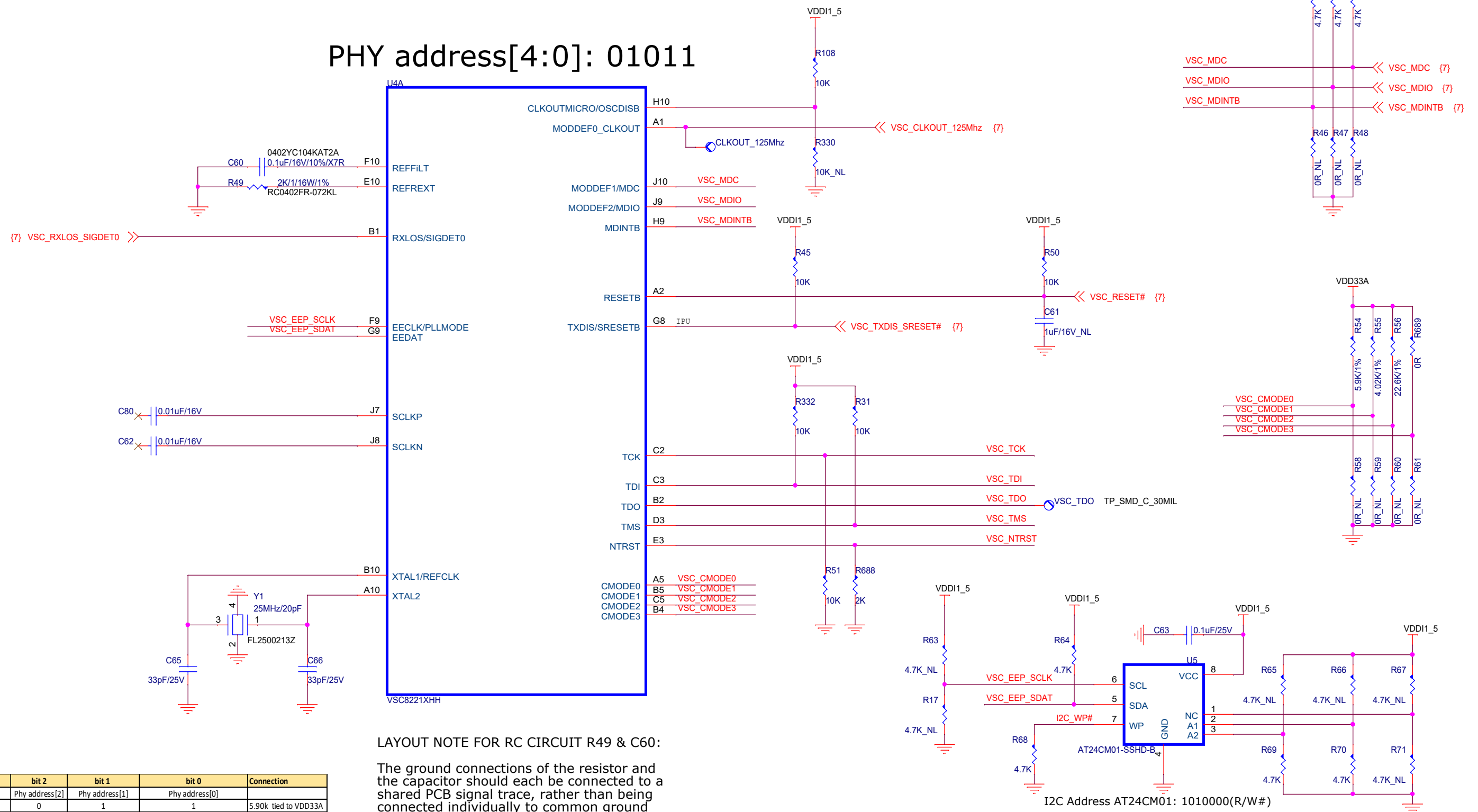


VSC8221-Power & Ground



VSC8221 - Connection

PHY address[4:0]: 01011



	bit 3	bit 2	bit 1	bit 0	Connection
CMODE 0	Phy address[3]	Phy address[2]	Phy address[1]	Phy address[0]	
	1	0	1	1	5.90k tied to VDD33A
CMODE1	SFP Mode Disable	PHY Address[4]	SIGDET pin direction	SerDes Line impedance	
	1	0	1	0	4.02k tied to VDD33A
CMODE 2	PHY Operating Mode[3]	PHY Operating Mode[2]	PHY Operating Mode[1]	PHY Operating Mode[0]	
	1	1	1	1	22.6k tied to VDD33A
CMODE3	LED Control[1]	SQE Enable	Reserved	Auto-negotiation Advertisement Control[1]	
	1	0	0	0	0k tied to VDD33A
	LED[2:0] = {Link/Activty, Link/Activty, Fault}			10/100/1000BASE-T HDX, FDX	

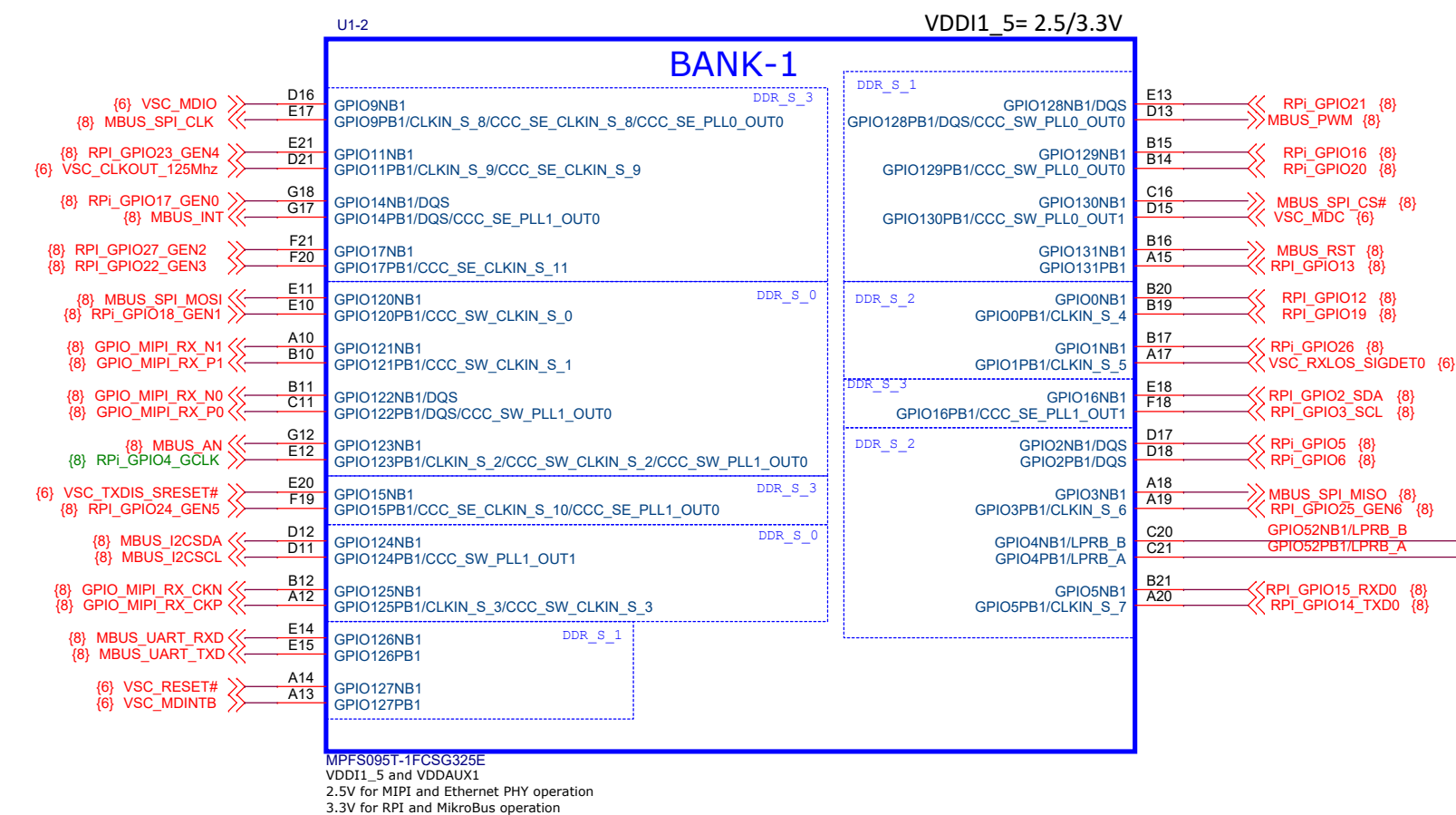
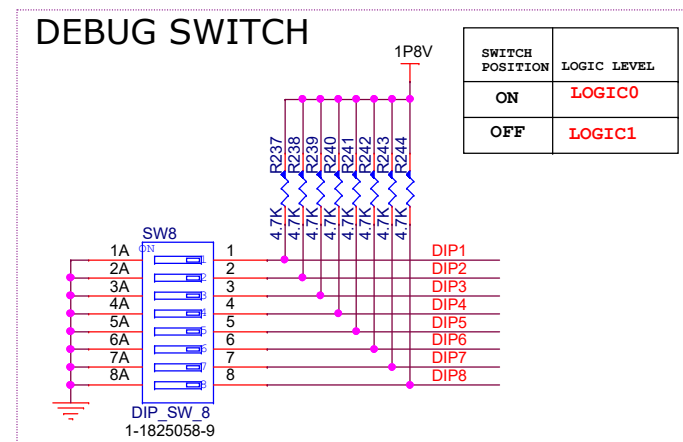
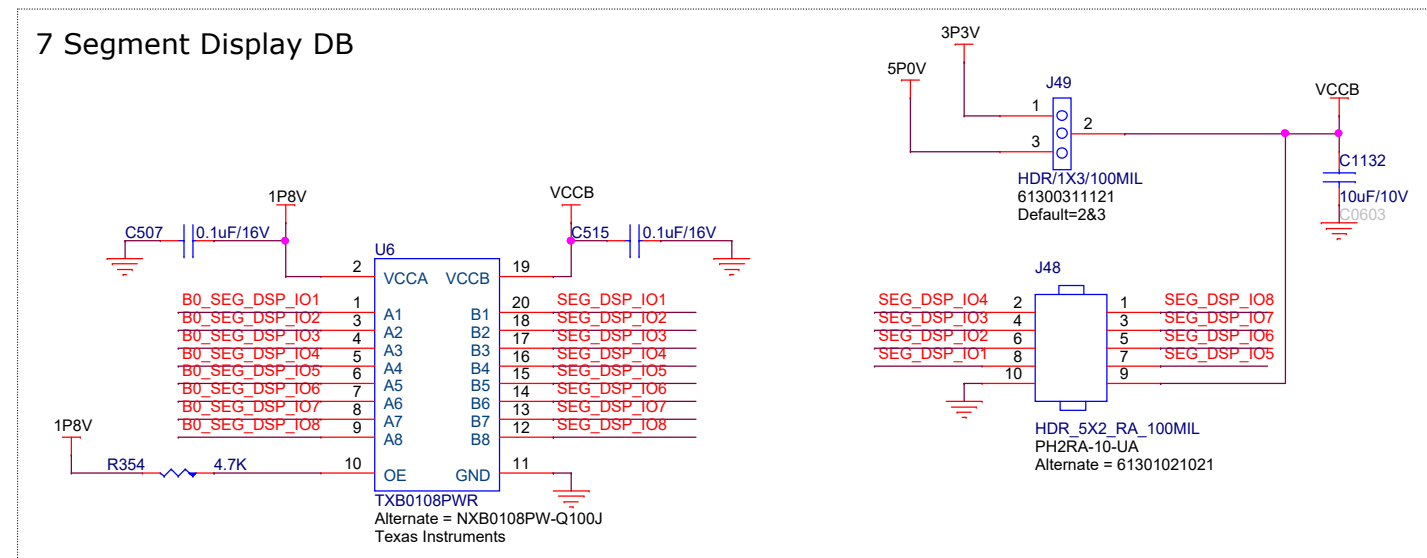
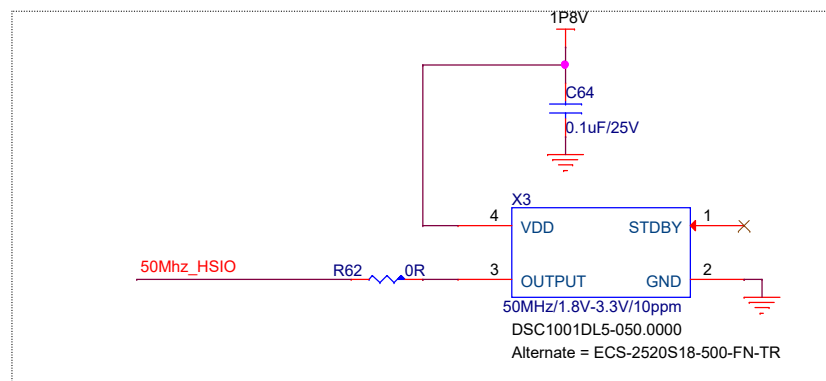
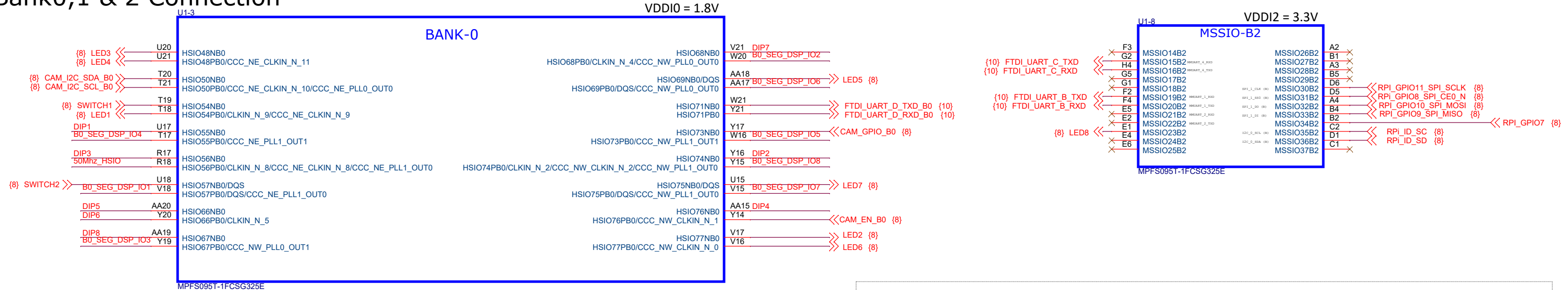
LAYOUT NOTE FOR RC CIRCUIT R49 & C60:

The ground connections of the resistor and the capacitor should each be connected to a shared PCB signal trace, rather than being connected individually to common ground plane. This PCB signal trace should then be connected to a ground plane at a single point. In addition, the reference capacitor and resistor should be placed as close as possible to the VSC8221

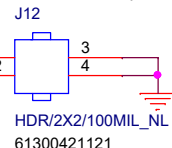


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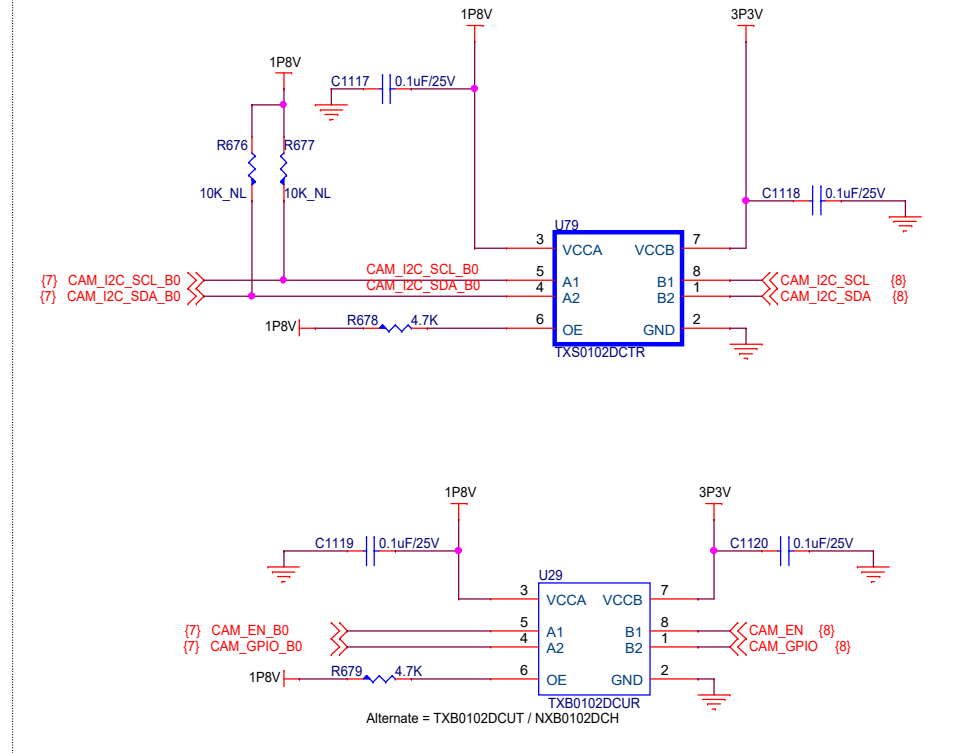
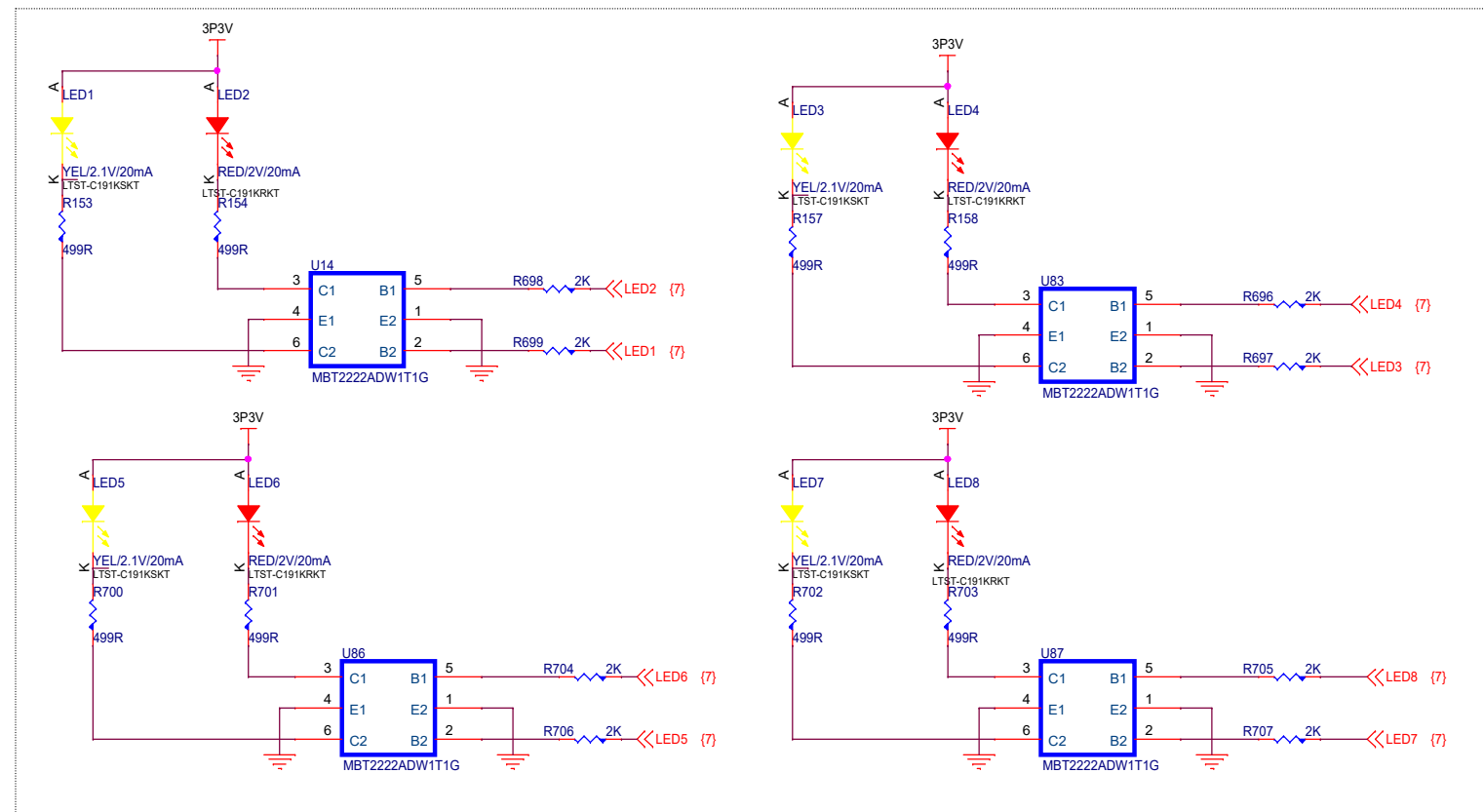
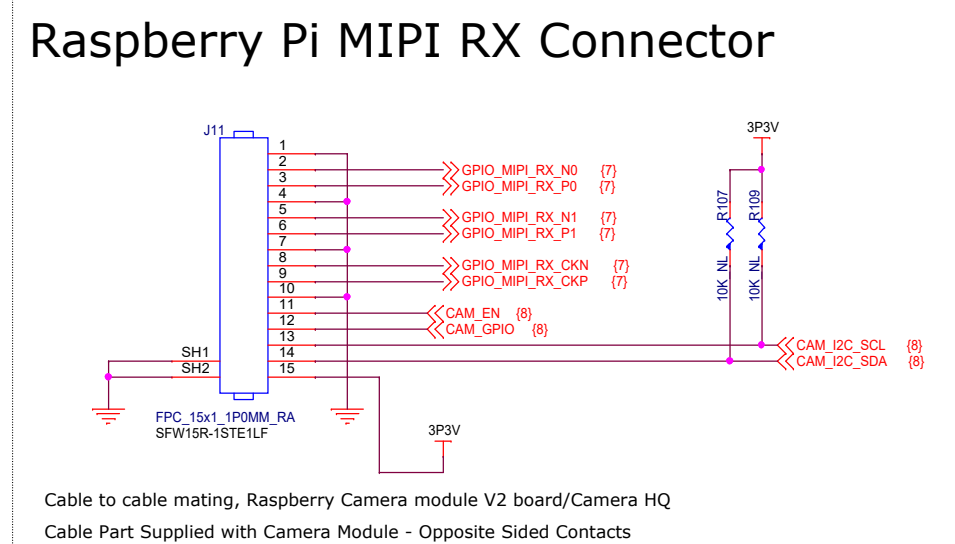
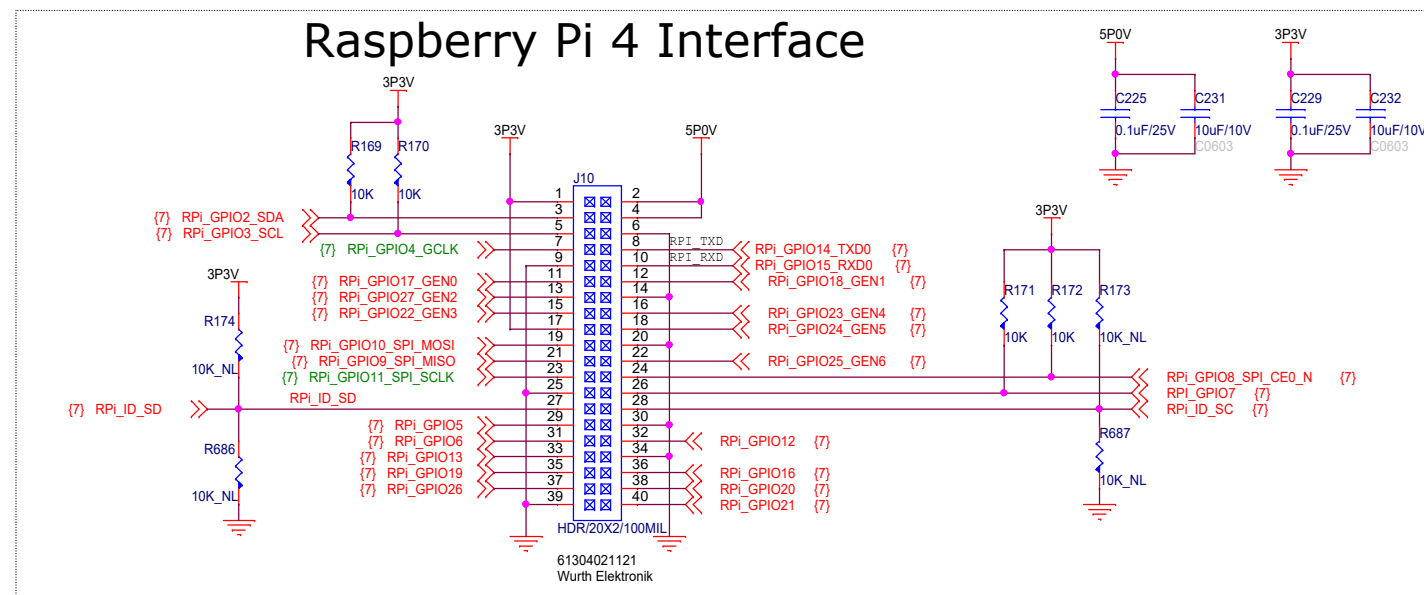
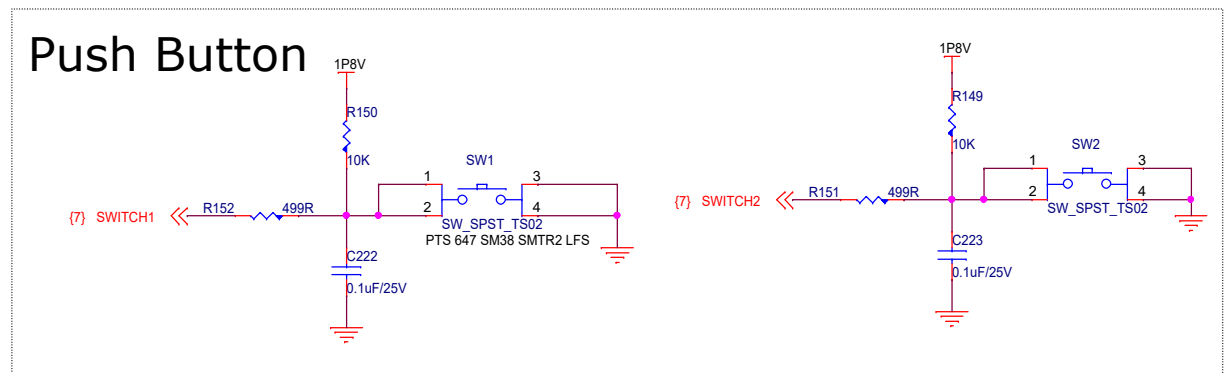
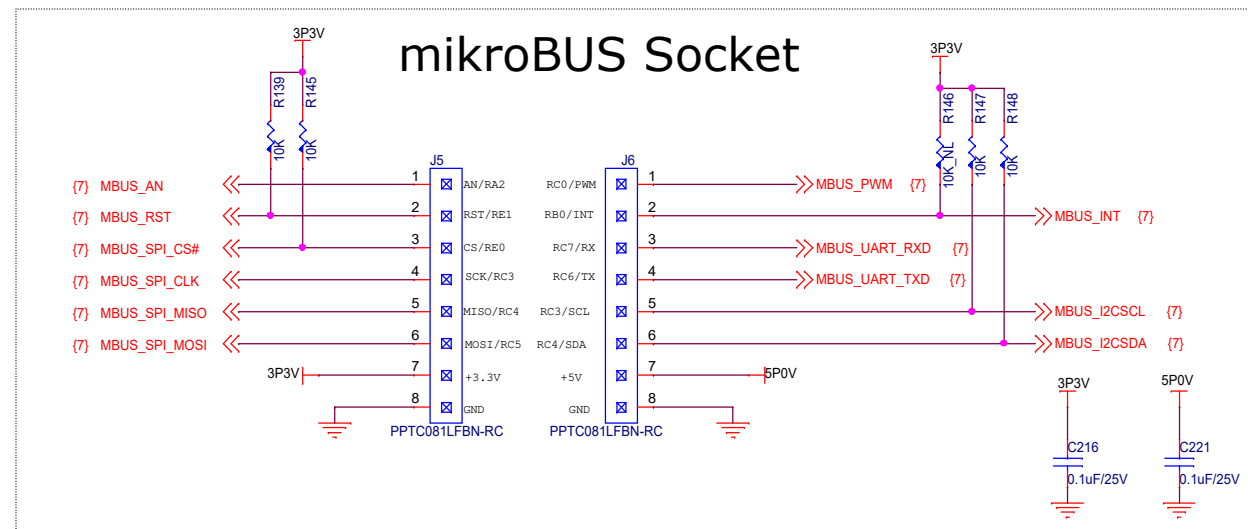
Bank0,1 & 2 Connection



NOTE:
Nets from pins C20 & C21
should be
routed differentially.

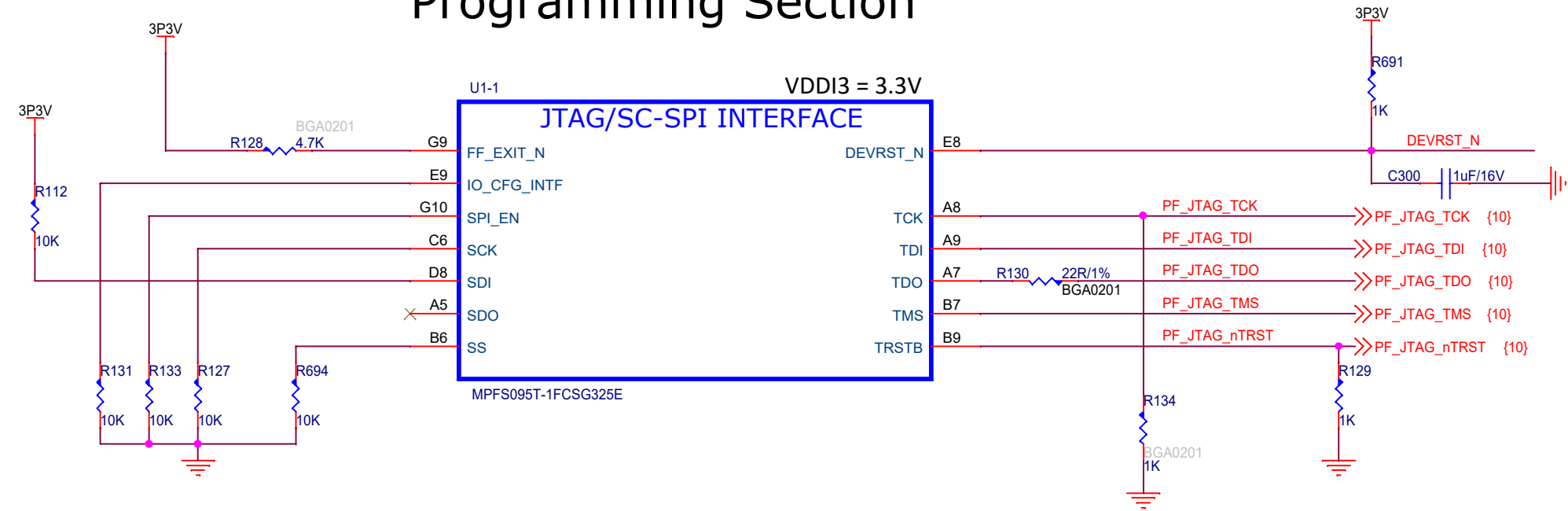


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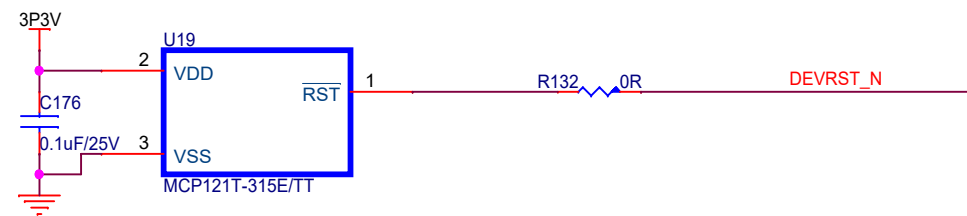


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Programming Section

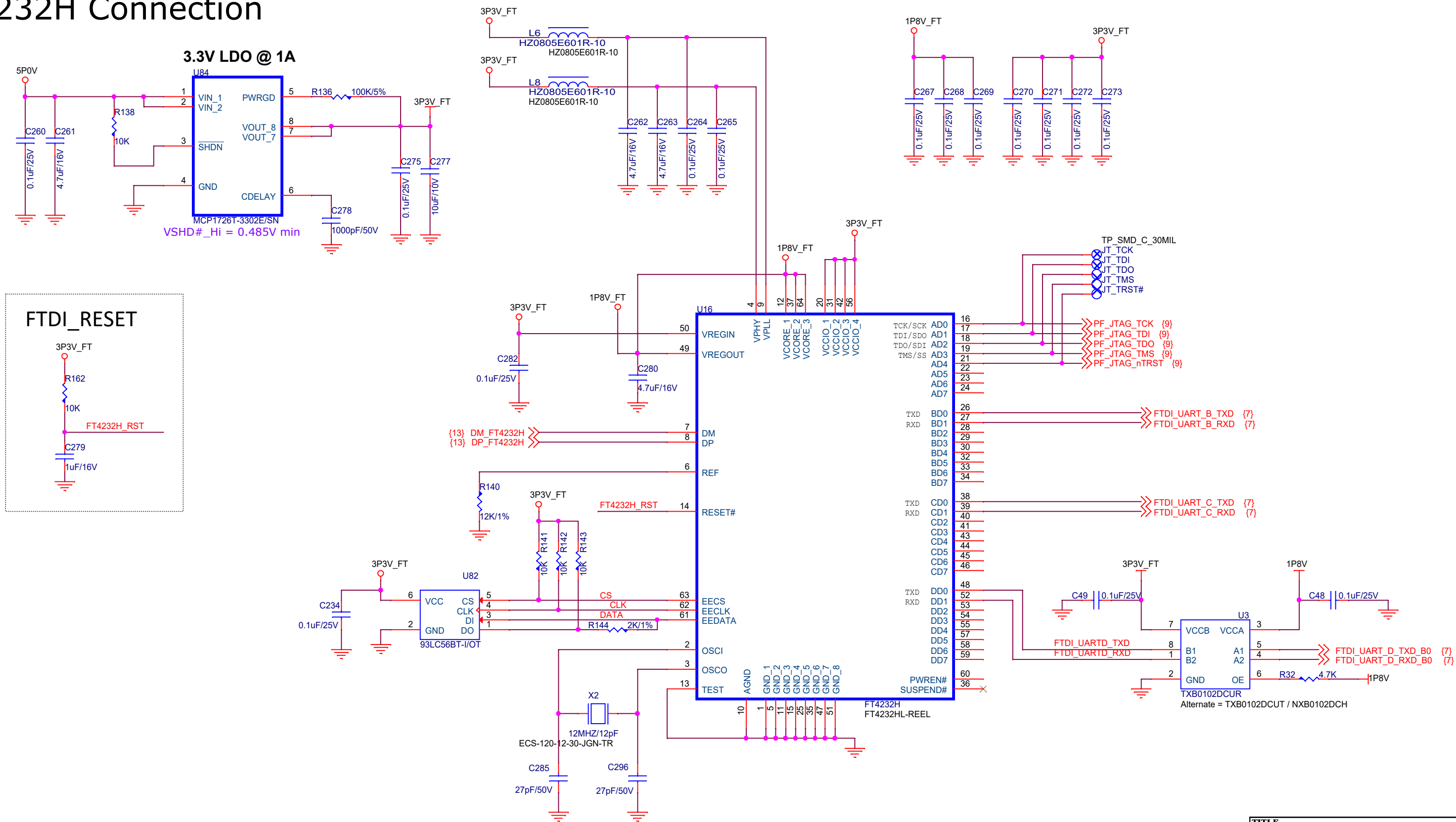


Device Reset



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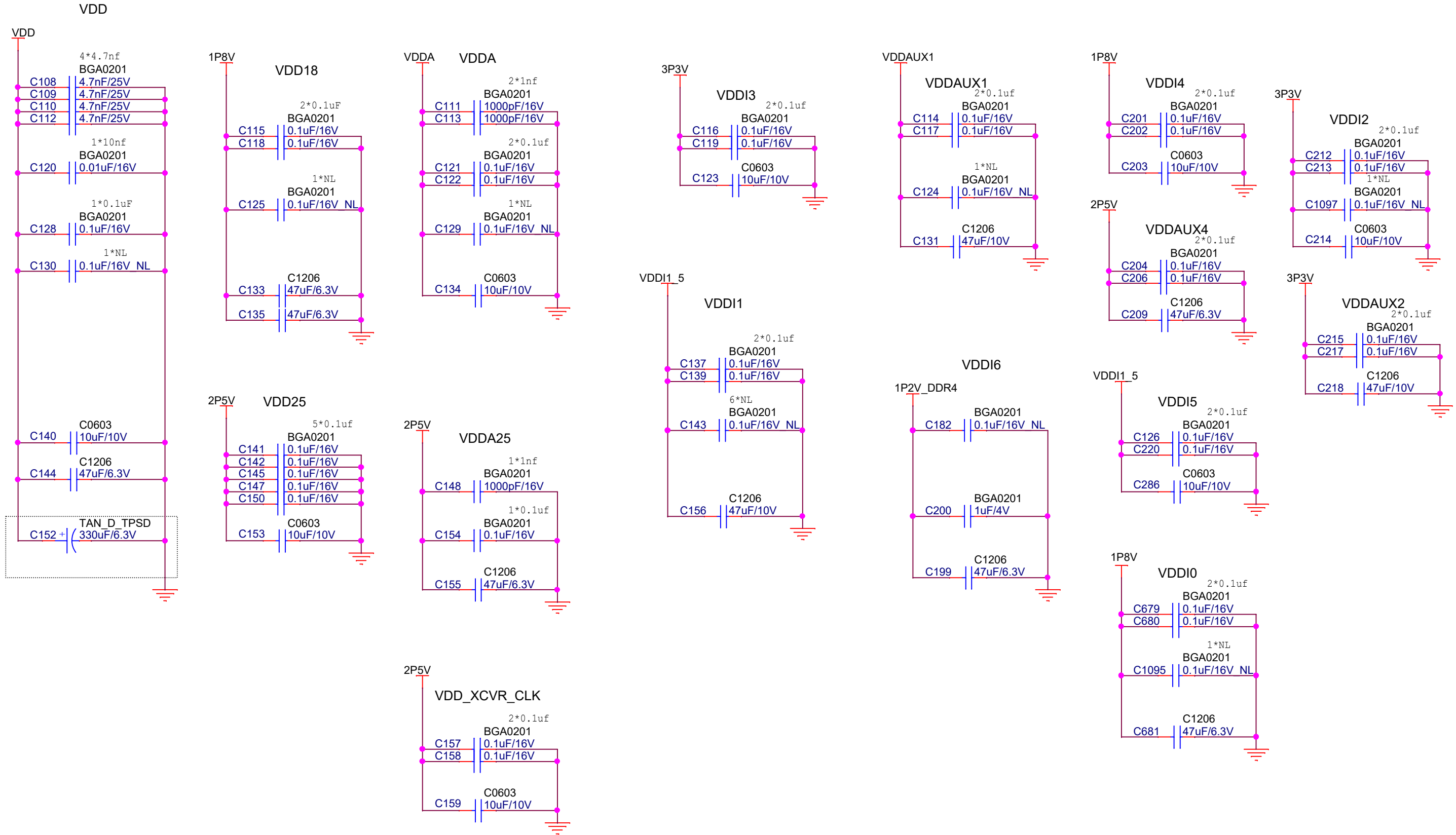
FT4232H Connection



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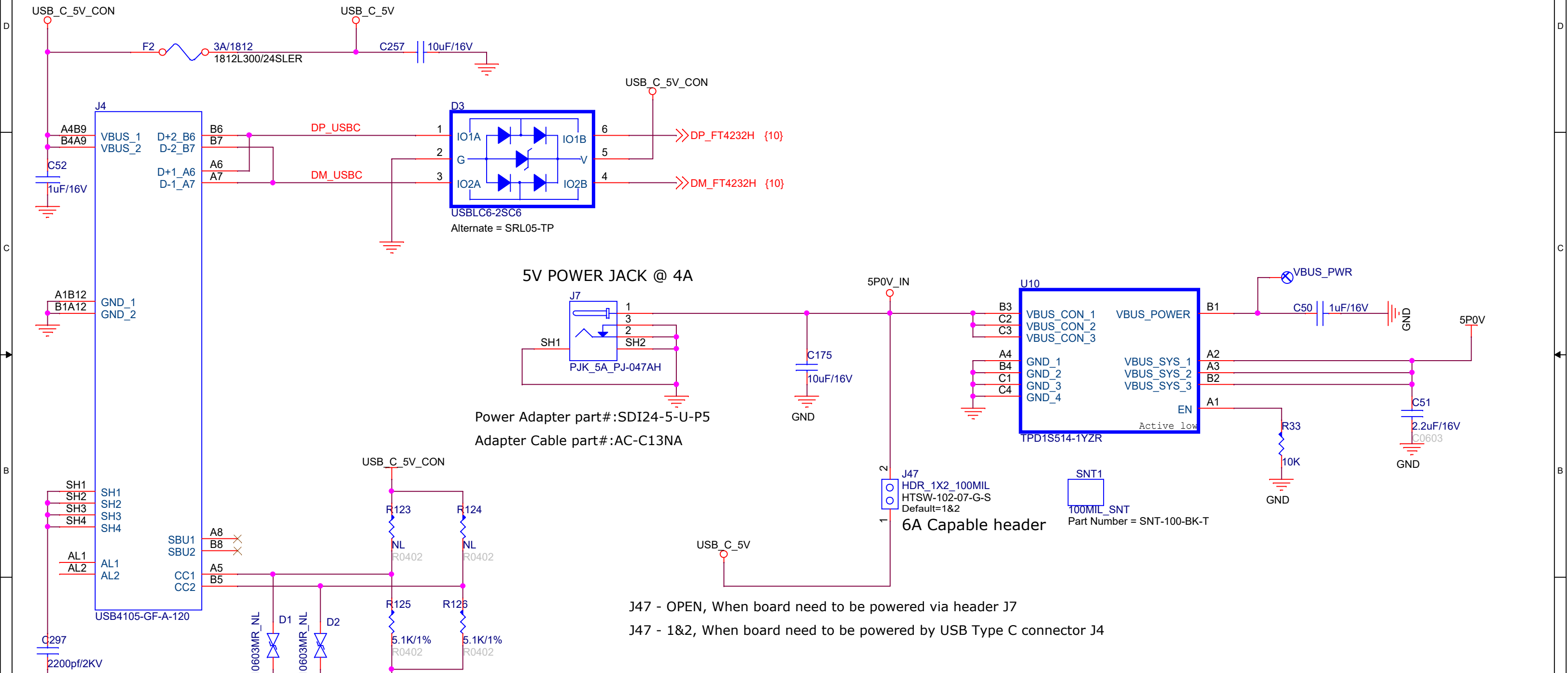


Decoupling Capacitors



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5V DC Supply



J47 - OPEN, When board need to be powered via header J7
J47 - 1&2, When board need to be powered by USB Type C connector J4



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